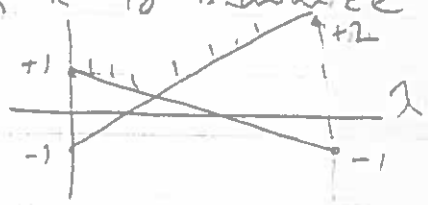
Game Theory Exam 2018
(Model Answers)

1(A): (a) If A chooses a mixed strategy $(1-\lambda), \lambda)$ A's pay-off is: $(1-\lambda)-\lambda$ if B chooses col. 1, and $-(1-\lambda)+2\lambda$ if B chooses col 1. So A's safety strategy is to minimize

$$\max\{1-2\lambda, -1+3\lambda\}$$



This is minimized at "cross-over" when

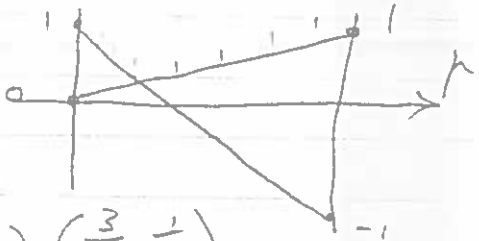
$$1-2\bar{\lambda} = -1+3\bar{\lambda} \Rightarrow \bar{\lambda} = 2/5$$

Likewise, a mixed safety strategy $((1-\bar{\mu}), \bar{\mu})$ minimizes

$$\max\{\mu, (1-\mu)-2\mu\}.$$

The cross-over condition gives

$$\bar{\mu} = 1-3\bar{\mu} \Rightarrow \bar{\mu} = \frac{1}{4}$$



Pay-off $(\frac{2}{5}, \frac{1}{4})$ The pair of mixed safety strategies is

$$(1-\bar{\lambda}, \bar{\lambda}), (1-\bar{\mu}, \bar{\mu}) = \left(\left(\tfrac{3}{5}, \tfrac{2}{5}\right), \left(\tfrac{3}{4}, \tfrac{1}{4}\right)\right)$$

We see $[(1-\lambda), \lambda] S^A \begin{bmatrix} 1-\bar{\mu} \\ \bar{\mu} \end{bmatrix} = \frac{1}{2}(1-\lambda) - \frac{1}{4}\lambda = \frac{1}{2} - \frac{3}{4}\lambda$

This is not minimized at $\bar{\lambda} = \frac{1}{2}$, so this not a Nash equilibrium.

(b) Since $((1-\lambda', \lambda'), (1-\mu', \mu'))$ is a Nash equilibrium in mixed strategies, we must have

① $[(1-\lambda), \lambda] S^A \begin{bmatrix} 1-\bar{\mu} \\ \bar{\mu} \end{bmatrix} \geq [(1-\lambda'), \lambda'] S^A \begin{bmatrix} 1-\mu' \\ \mu' \end{bmatrix}$ for all λ

② $[(1-\lambda'), \lambda'] S^B \begin{bmatrix} 1-\mu \\ \mu \end{bmatrix} \geq [(1-\lambda', \lambda'] S^B \begin{bmatrix} 1-\mu' \\ \mu' \end{bmatrix}$ for all μ

① $\Rightarrow [(1-\lambda), \lambda] \begin{bmatrix} (1-\mu')-\mu' \\ -(1-\mu')+2\mu' \end{bmatrix} = (1-\lambda)(1-2\mu') - \lambda(-1+3\mu')$ is minimized at $\lambda = \lambda'$.

Because λ' is an interior minimizer slope must be zero $\Rightarrow \mu' = \frac{2}{5}$

② $\Rightarrow [\lambda', (1-\lambda')-2\lambda'] \begin{bmatrix} 1-\mu \\ \mu \end{bmatrix} = \lambda'(1-\mu) + (1-3\lambda')\mu$ is minimized at $\mu = \mu'$

Pay-off $(\frac{1}{5}, \frac{1}{4})$ Since the minimizer is interior, slope is zero $\Rightarrow \lambda' = 1/4$

The Nash equilibrium pay-offs are not worse than the safety pay-offs, as you would expect since Nash equilibria take account of the other player's rational behaviour.

(B): Take any pair of strategies (x, y). Then

$$\text{Min}_{x'} L(x', y) \leq L(x, y) \leq \text{Max}_{y'} L(x, y')$$

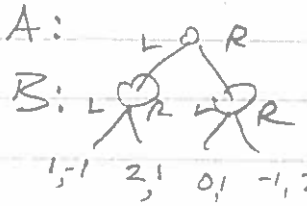
Since the right side does not depend on y

$$\text{Max}_{y'}(\text{Min}_{x'} L(x', y') \leq \text{Max}_{y'} L(x, y')$$

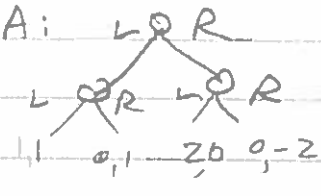
Since the left side does not depend on x,

$$\text{Max}_{y'}(\text{Min}_{x'} L(x, y')) \leq \text{Min}_{x'}(\text{Max}_{y'} L(x, y'))$$

2. The trees associated with the 2 chance scenarios are



and



The strategies for A are (L, R)

B's strategies are $\binom{B_1}{L}, \binom{B_2}{R}, \binom{B_3}{R}, \binom{B_4}{L}$ when $A = \binom{L}{R}$

The pay-off matrices are obtained from the paths

A \ B	B_1	B_2	B_3	B_4
L	LL	LR	LR	LL
R	RR	RL	RR	RL

They are

scenario ① :

A \ B	B_1	B_2	B_3	B_4
L	(1,-1)	(2,1)	(2,1)	(1,-1)
R	(-1,2)	(0,1)	(-1,2)	(0,1)

and scenario ② :

A \ B	B_1	B_2	B_3	B_4
L	(1,1)	(0,1)	(0,1)	(1,1)
R	(0,-2)	(2,0)	(0,-2)	(2,0)

Mixing them $\frac{1}{2} \times \{\text{scenario 1}\} + \frac{1}{2} \times \{\text{scenario 2}\}$ gives

A \ B	B_1	B_2	B_3	B_4
L	(1,0)	(1,1)	(1,1)	(1,0)
R	$(-\frac{1}{2},0)$	$(1,\frac{1}{2})$	$(-\frac{1}{2},0)$	$(1,\frac{1}{2})$

The Nash equilibria are (L, B_4) (R, B_1), (R, B_3)

These give (L, L) (pay-off (1,0)), (R, R) (pay-off (-1,2))

(Notice $(R B_3)$ duplicates (R, B_1) and gives (R, R))

(L, B_4) is weakly dominated, so not admissible

(R, B_1) $(\equiv (R B_3))$ is not weakly dominated, so is admissib

3. Assume the ith juror votes 'not guilty' w.p. α and all other $(N-1)$ jurors vote "not guilty" w.p. $\bar{\alpha}$. $(\alpha, \bar{\alpha} \in (0,1))$

We must calculate

A = Prob{juror i votes 'not guilty' and 0 or 1 other jurors vote 'not guilty'}

B = Prob{juror i votes 'guilty' and 0, 1 or 2 other jurors vote 'not guilty'}

But

A = Prob{juror i votes 'not guilty'} × (Prob{0 other jurors vote not guilty} + Prob{exactly 1 other juror votes 'not guilty'}

$= \alpha \left(P_{\bar{\alpha}}(0; N-1) + P_{\bar{\alpha}}(1; N-1) \right)$

B = Prob{juror i votes guilty}

× (Prob{0 jurors vote 'not guilty'} + Prob{1 ...} + Prob{2..}

$= (1-\alpha)\left(P_{\bar{\alpha}}(0; N-1) + P_{\bar{\alpha}}(1; N-1) + P_{\bar{\alpha}}(2; N-1) \right)$

The pay-off is $A + (1-c)B$

$J(\alpha; \bar{\alpha}) := \alpha \left(P_{\bar{\alpha}}(0; N-1) + P_{\bar{\alpha}}(1; N-1) \right)$

$+ (1-c)(1-\alpha)\left(P_{\bar{\alpha}}(0; N-1) + P_{\bar{\alpha}}(1; N-1) + P_{\bar{\alpha}}(2; N-1) \right)$

If $(\bar{\alpha}, \ldots, \bar{\alpha})$ is a mixed Nash equilibrium, we require then

$$J(\alpha; \bar{\alpha}) \leq J(\bar{\alpha}; \bar{\alpha})$$

$\alpha \to J(\alpha; \bar{\alpha})$ is linear. So this is only possible if 'slope = 0', i.e

$$P_{\bar{\alpha}}(0; N-1) + P_{\bar{\alpha}}(1; N-1) - (1-c)\left(P_{\bar{\alpha}}(0; N-1) + P_{\bar{\alpha}}(1; N-1) + P_{\bar{\alpha}}(2; N-1) \right) = 0$$

This gives

$$\left(P_{\bar{\alpha}}(0; N-1) + P_{\bar{\alpha}}(1; N-1) + P_{\bar{\alpha}}(2; N-1) \right) \times c = P_{\alpha}(2; N-1)$$

Using data $c = \dfrac{\frac{(N-1)(N-2)}{2}\, \bar{\alpha}^2 (1-\bar{\alpha})^{N-3}}{(1-\bar{\alpha})^{N-1} + (N-1)\bar{\alpha}(1-\bar{\alpha})^{N-2} + \frac{1}{2}(N-1)(N-2)\bar{\alpha}^2 (1-\bar{\alpha})^{N-3}}$

We see the right is 0 when $\bar{\alpha} = 0$. The 'by continuity',

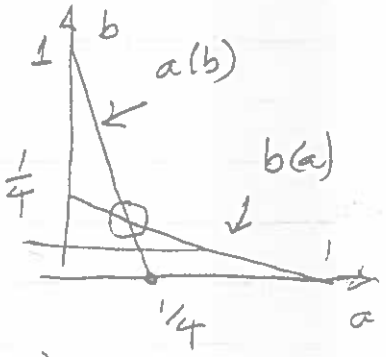
$\bar{\alpha} \to 0$ as $c \to 0$,

4 (a) The pay-offs are $L^A(a,b) = a(1-[a+b]) - a^2$
and $L^B(a,b) = b(1-[a+b]) - b^2$
If $(\bar{a},\bar{b})$ is a Nash equilibrium for which $\bar{a},\bar{b} > 0$,
then, $\partial/\partial a\, L^A(\bar{a},\bar{b}) = 0$ and $\partial/\partial b\, L^B(\bar{a},\bar{b}) = 0$.
This gives

$$(1-[\bar{a}+\bar{b}]) - \bar{a} - 2\bar{a} = 0 \Rightarrow \bar{a} = \tfrac{1}{4}(1-\bar{b})$$

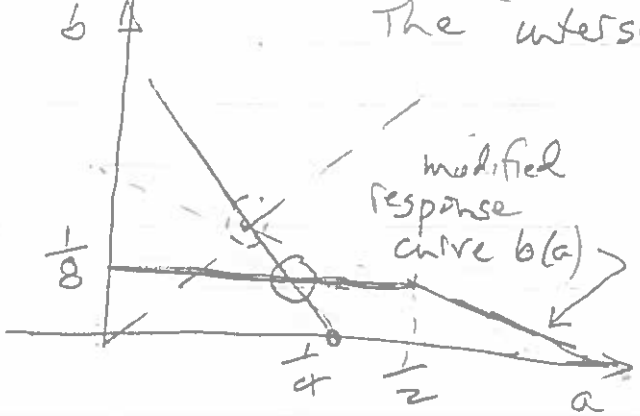
By symmetry $\bar{b} = \frac{1}{4}(1-\bar{a})$.
Hence $\bar{a} = \frac{1}{4}(1-\frac{1}{4}(1-\bar{a})) \Rightarrow \underline{\bar{a} = \tfrac{1}{5}}$ and $\underline{\bar{b} = \tfrac{1}{5}}$ (by symmetry)

(b) The response curves for company A and B resp are given by $\partial/\partial a\, L^A(a=a(b),b) = 0 \Rightarrow a(b) = \frac{1}{4}(1-b)$
and also $b(a) = \frac{1}{4}(1-a)$
The intersection is at the solution of
$a = \frac{1}{4}(1-b)$ and $b = \frac{1}{4}(1-a) \Rightarrow (\frac{1}{5},\frac{1}{5})$.
This is consistent with the calculated Nash equilibrium $(\bar{a},\bar{b}) = (\frac{1}{5},\frac{1}{5})$



(c) The Stackelberg optimizer maximizes
$a \to L^A(a, b(a)) = a - 2a^2 - \frac{1}{4}(1-a)$
$\partial/\partial a(..) = 0 \Rightarrow 5/4 - 4\tilde{a} = 0 \Rightarrow \underline{\tilde{a} = \tfrac{5}{16}}$
B's price is then $\tilde{b} = \frac{1}{4}(1-\tilde{a}) \Rightarrow \underline{\tilde{b} = 11/32}$

(d) Assume B production is constrained by $0 < b \le \frac{1}{8}$.
Then B's response curve becomes
$b_{\text{modified}}(a) = \max\{\frac{1}{4}(1-a), \frac{1}{8}\}$
Noting the 'cut-off' occurs when $\frac{1}{4}(1-a') = \frac{1}{8}$
We can now draw the modified response curve $b_{\text{modified}}(a)$. ($a(b)$ remains the same.)
$\Rightarrow a' = 1/2$



The intersection of response curves is now $a' = \frac{1}{4}(1-\frac{1}{8})$, $b' = \frac{1}{8}$
Modified Nash equilibrium is $(\frac{7}{32}, \frac{1}{8})$

5A The Bellman equation is $V_t(x) = \min_u \{ V_{t+1}(Ax+bu) + u^2 \mid u \in \mathbb{R} \}$

with boundary condition $V_N(x) = \bar{\lambda}' x$ (*)

Assume $V_t(x) = x^T P_t x + 2 r_t^T x + s_t$

Then $x^T P_t x + 2 r_t x + s = \min_u \{ (Ax+bu)^T P_{t+1} (Ax+bu) + 2 r_t^T (Ax+bu) + u^2 \}$

$= \min_u \{ x^T A^T P_{t+1} A x + 2 (x^T A^T P_{t+1} b + r_{t+1}^T b) u + 2 r_{t+1}^T A x + (b^T P_{t+1} b + 1) u^2 \}$

The minimizing $u = -(b^T P_{t+1} b + 1)^{-1} (b^T P_{t+1} A x + b^T r_{t+1})$

Plug this into Bellman equation:

$x^T P_t x + 2 r_t^T x =$

$x^T (A^T P_{t+1} A + 2 r_{t+1} A x - (x^T A^T P_{t+1} b + r_{t+1}^T b)^2 (b^T P_{t+1} b + 1)^{-1}$

Equating 1st and 2nd powers of x on both sides gives:

$P_t = A^T P_{t+1} A - (b^T P_{t+1} b + 1)^{-1} A^T P_{t+1} b b^T P_{t+1} A$ — (1)

$r_t = A^T r_{t+1} - (b^T P_{t+1} b + 1)^{-1} A^T P_{t+1} b b^T r_{t+1}$ — (2)

$s_t = (b^T P_{t+1} b + 1)^{-1} r_{t+1}^T b b^T r_{t+1}$ — (3)

The boundary conditions are

$P_N = 0$, $r_N = \lambda$ and $s_N = 0$

The optimal control $\{u_t^\lambda\}$ (and state $\{x_t^\lambda\}$ are given by the feedback relation

$u_t = -k_t^T x_t - d_t^T$

$k_t = (b^T P_{t+1} b + 1)^{-1} b^T P_{t+1} A$, $d_t = (b^T P_{t+1} b + 1)^{-1} b^T r_{t+1}$

5B Suppose $\bar{\lambda}$ is such that, for the given initial state x_0, $x_N^{\bar{\lambda}} = 0$.

Take any control/state pair $\{x_t, u_t\}$ such that $x_N = 0$. Then, because $\{x_t^{\bar{\lambda}}, u_t^{\bar{\lambda}}\}$ is optimal for $\lambda = \bar{\lambda}$

$$\sum_{t=0}^{N-1} u_t^2 + \bar{\lambda}^T x_N \geq \sum_{t=0}^{N-1} |u_t^{\bar{\lambda}}|^2 + \bar{\lambda}^T x_N^{\bar{\lambda}}.$$

Since $x_N = 0$ and $x_N^{\bar{\lambda}} = 0$, it follows

$$\sum_{t=0}^{N-1} u_t^2 \geq \sum_{t=0}^{N-1} |u_t^{\bar{\lambda}}|^2$$

This inequality tells us that $\{x_t^{\bar{\lambda}}, u_t^{\bar{\lambda}}\}$ minimizes the control energy for transferring the state from x_0 to 0